

Chapter 1. Introduction

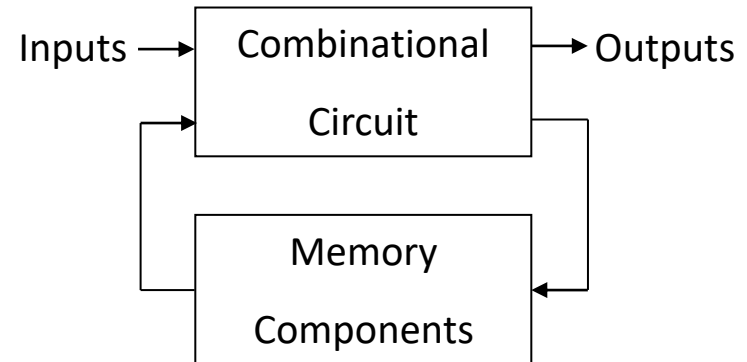
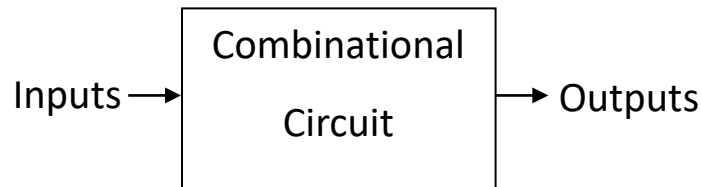
Latches ve Flip-Flops

Flip-Flop Types

- SR type
- D type
- T type
- JK type

Introduction

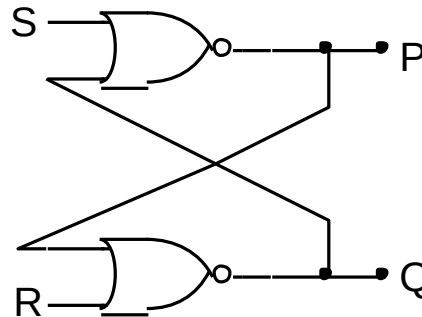
Logic circuits are divided into two categories: Combinational circuits and sequential circuits. Sequential circuits have two forms: Synchronous and asynchronous. Outputs of combinational circuits depend on the current value of the inputs. On the other hand, outputs of sequential circuits depend on the current and the previous value of the inputs. Latches, flip-flops and registers are used as memory components in sequential circuits.



Latches and Flip-Flops

Latches are memory components which store binary data and are formed by connecting two logic gates in a form of positive feedback.

SR Type Latch:



$$P = (S+Q)'$$
$$Q = (R+P)'$$

- **Store State:** This is the state where both inputs are zeros.

When $S = R = 0$, the outputs are complements of each other. $P = Q'$ and $Q = P'$.

- **Loading Data:** A latch stores either 1 or 0 as a value.

⇒ To store 1, inputs are set as $S = 1$ and $R = 0$. In this case $Q = 1$ and $P = 0$

⇒ To store 0, inputs are set as $S = 0$ and $R = 1$. In this case $Q = 0$ and $P = 1$

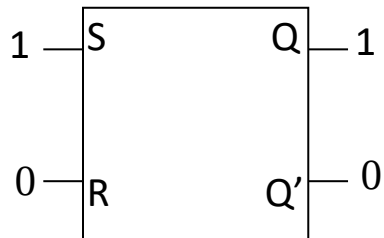
The output P can be also named as Q' .

The inputs S and R stands for Set and Reset.

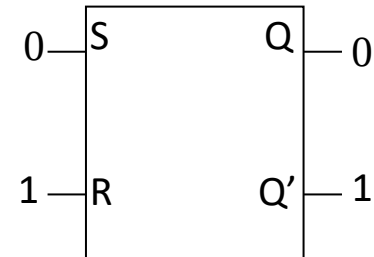
SR Type Latch

To conclude on diagram;

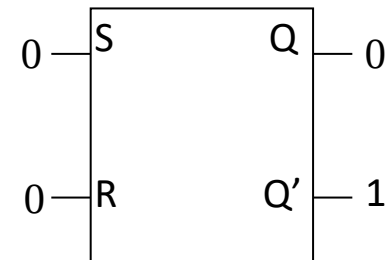
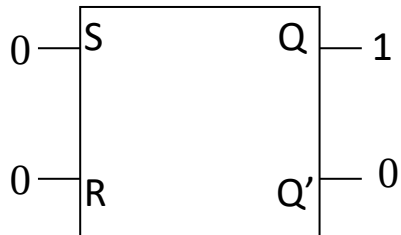
Set State



Reset State



If we set $SR = 00$ afterwards, the latch goes into store state.



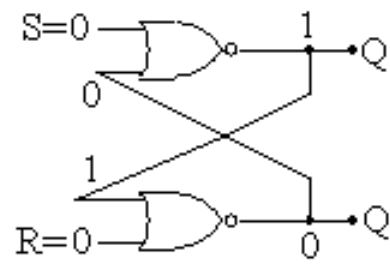
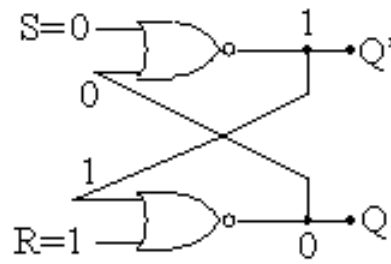
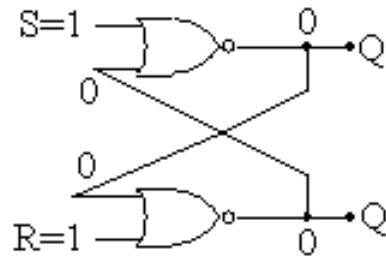
SR Type Latch

If both inputs are set to 1;

We apply $SR = 11$ in definition equations: $P = (S+Q)'$ and $Q = (R+P)'$

$P = (1+Q)' = 0$ and $Q = (1+P)' = 0$

Both outputs are 0 and they are not compliments of each other. This is invalid.
So SR latch doesn't allow both inputs to be set to 1 at the same time.



Flip-Flops (F/F)

Flip-flops are memory components which have clock signal input.

A flip-flop's output value changes only when the clock signal changes.

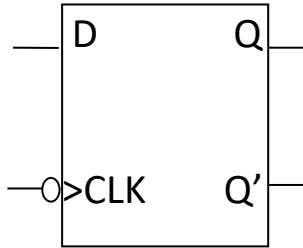
Flip flops which are triggered when clock input changes from 1 to 0 (high to low) are called negative-edge triggered, and the ones which are triggered when clock input changes from 0 to 1 (low to high) is called positive-edge triggered.

What to store in a flip flop depends on data inputs and the data stored in previous clock signal.

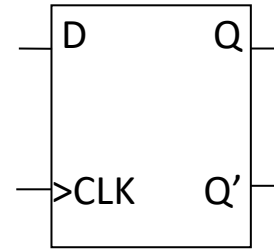
Basically, there are 4 types of flip-flops D, JK, SR and T.

D Type Flip-Flop

D stands for Delay. The input is delayed until next clock pulse. The next output of D flip-flop depends on the D value before the clock pulse.



Negative-edge triggered



Positive-edge triggered

Truth table of D type F/F

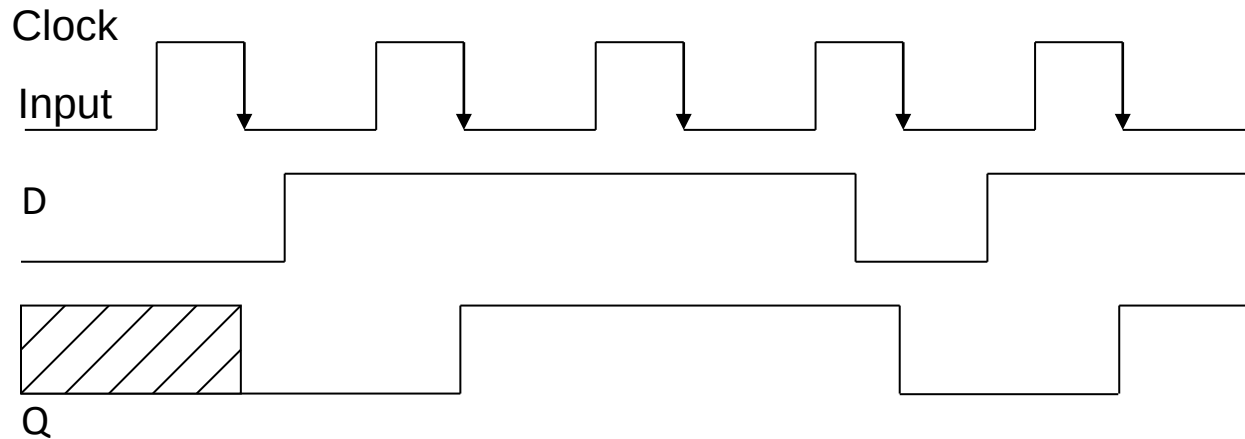
D	q	Q
0	0	0
0	1	0
1	0	1
1	1	1

D	Q
0	0
1	1

$$Q = D$$

D Type Flip-Flop

Let's analyze the behavior of negative-edge triggered D type F/F on time diagram.



SR Type Flip-Flop

SR type F/F has two inputs called Set and Reset just like SR latch.

Truth table of SR type F/F

S	R	q	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	-
1	1	1	-

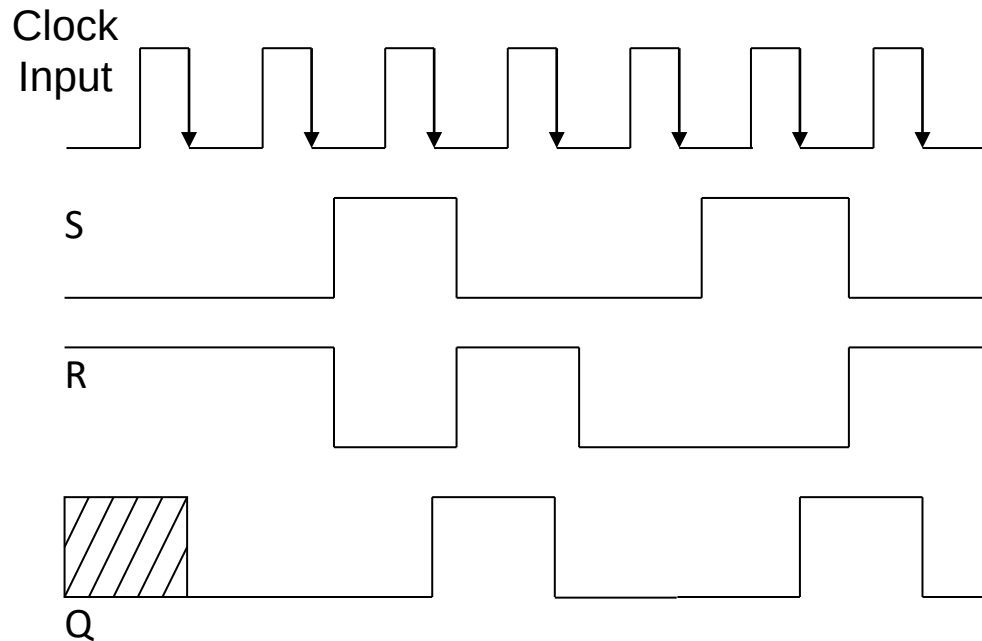
S	R	Q
0	0	q
0	1	0
1	0	1
1	1	-

		Q			
q \ SR		00	01	11	10
				x	1
0				x	1
1		1		x	1

$$Q = S + R'.q$$

SR Type Flip-Flop

Let's analyze the behavior of negative-edge triggered SR type F/F on time diagram.



T Type Flip-Flop

T type F/F has an input called T (Toggle). If $T=1$, flip-flop complements its output value, and if $T=0$ it keeps the current output value.

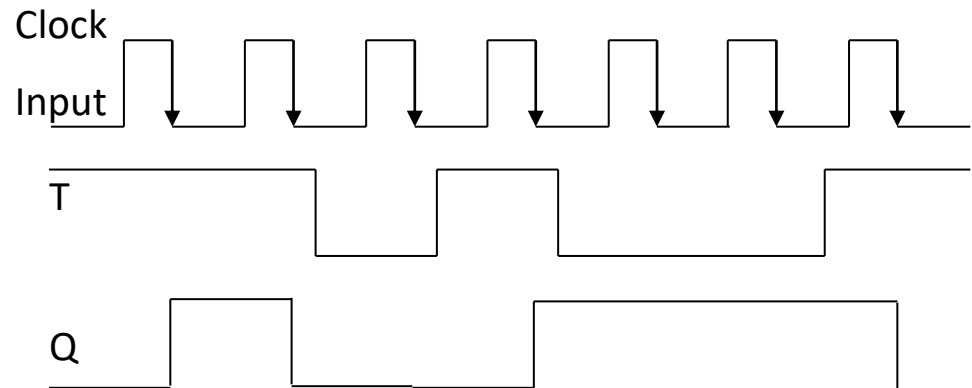
Truth table of T type F/F

T	q	Q
0	0	0
0	1	1
1	0	1
1	1	0

T	Q
0	q
1	q'

$$Q = T \oplus q$$

Let's analyze the behavior of negative-edge triggered T type F/F on time diagram.



JK Type Flip-Flop

JK type F/F has two inputs called J and K. It can be considered as combination of SR and T type flip-flops. It behaves as SR flip flop except J=K=1 situation. In case of J=K=1 it behaves as T flip flop.

Truth table of JK type F/F

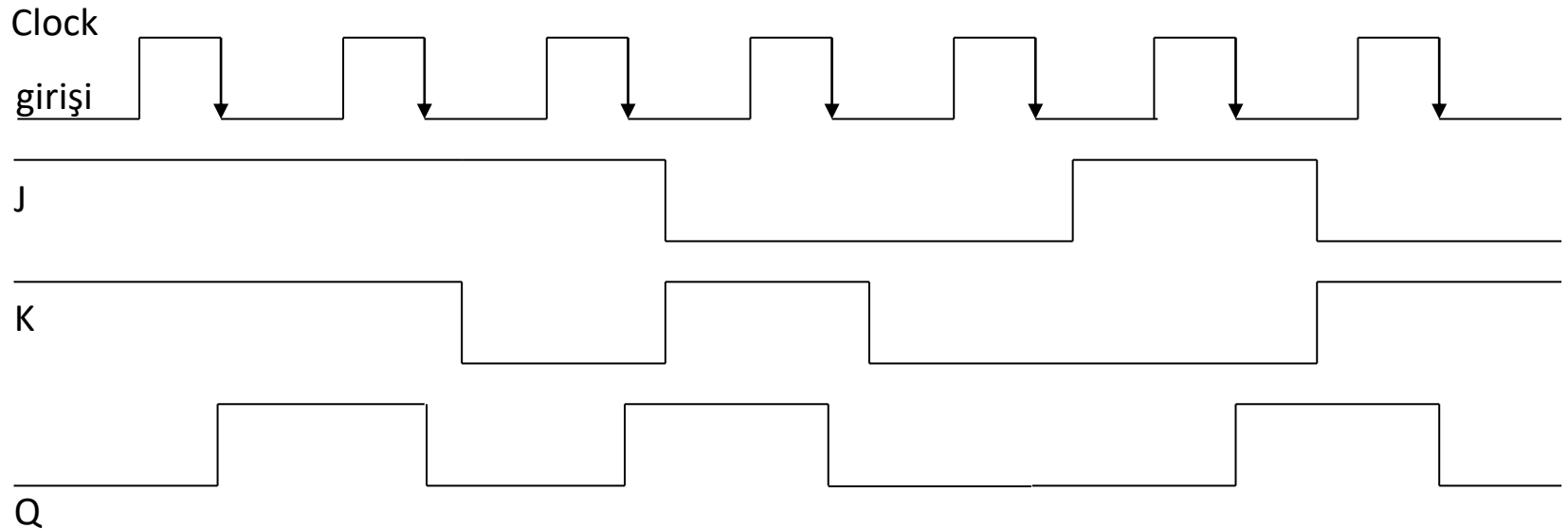
J	K	q	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

J	K	Q
0	0	q
0	1	0
1	0	1
1	1	q'

$$\begin{aligned} Q &= J'K'q + JK' + JKq' \\ &= J'K'q + JK'(q+q') + JKq' \\ &= J'K'q + JK'q + \underline{JK'q'} + JKq' \\ &= \underline{J.q'} + K'.q \text{ (Characteristic equation)} \end{aligned}$$

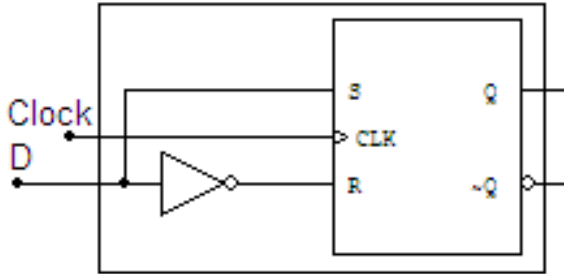
JK Type Flip-Flop

Let's analyze the behavior of negative-edge triggered D type F/F on time diagram and consider initial value of $Q=0$;



Conversion of Flip-Flops

⇒ We can convert SR and JK flip-flops to D flip flop;

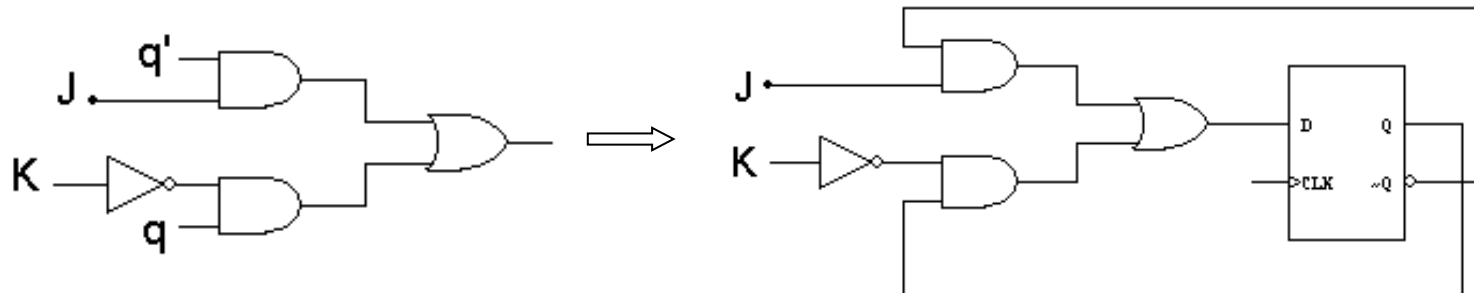


$$Q = S + R'.q$$

$$Q = D(q+1) = D + D'q$$

⇒ Characteristic equation of JK flip-flop is, $Q = J.q' + K'.q$

If we input this logic statement to D flip-flop, we get JK flip-flop;



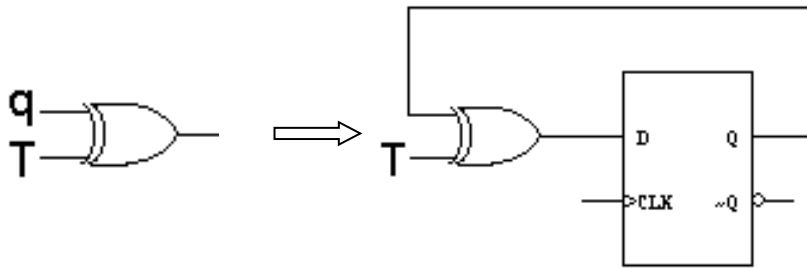
$$Q = D$$

$$Q = J.q' + K'.q$$

Conversion of Flip-Flops

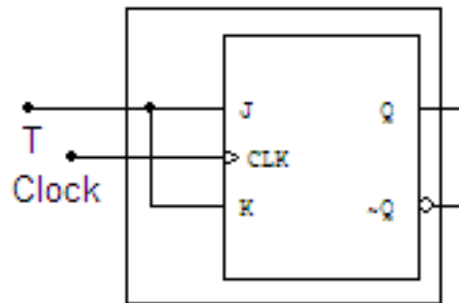
⇒ Characteristic equation of T flip-flop is $Q = T \oplus q$

If we input this logic statement to D flip-flop, we get T flip flop;



$$Q = D$$
$$Q = T \oplus q$$

⇒ If we connect two inputs of JK flip-flop, we get T flip-flop;



$$Q = T \oplus q = T.q' + T'.q$$
$$Q = J.q' + K'.q$$